

AMCS2034 Introduction to Data Structures and Algorithms

Tutorial 7: Types of Data Structures (PART A)

- Q1. What is a static data structure?
- Q2. Give **ONE (1)** example of a static data structure.
- Q3. What is the appropriate usage of a static data structure?
- Q4. Explain the motivation for using a static data structure even though its maximum size needs to be set in advance.
- Q5. What is a dynamic data structure?
- Q6. State **ONE (1)** example of a dynamic data structure.
- Q7. Provide a reason when a dynamic data structure should be used.
- Q8. Explain a potential disadvantage when using a dynamic data structure.
- Q9. Describe any **THREE (3)** differences between an array and a linked list.
- Q10. State **TWO (2)** advantages that a linked list has over an array.
- Q10. Describe **THREE (3)** disadvantages of linked list.

Tutorial 7: Types of Data Structures (PART B)

- Q12. Describe linear data structures.
- Q13. Briefly explain the following linear data structures.
- (i) Stack
 - (ii) Queue
- Q14. Describe the **THREE (3)** operations for a queue.
- Q15. Explain **TWO (2)** ways that a queue can be represented for implementation.
- Q16. Briefly explain the **FOUR (4)** operations for a stack.
- Q17. Explain **TWO (2)** ways that a stack can be represented for implementation.
- Q18. Brief explain the following **TWO (2)** nonlinear data structures.
- (i) Tree
 - (ii) Graph
- Q19. Describe each of the following binary tree traversal methods.
- (i) Preorder Traversal
 - (ii) Inorder Traversal
 - (iii) Postorder Traversal